

Appendix

A.1 Further Analysis

Covariates Type	Model	Covariate(s)	Algorithm	ACC		SEN		SPE	
				Train	Test	Train	Test	Train	Test
Functional	4.A	CCA	BG	0.891 (0.882,0.901)	0.892 (0.876,0.912)	0.97 (0.966,0.974)	0.971 (0.96,0.981)	0.26 (0.2,0.312)	0.275 (0.188,0.357)
	4.B		RF	0.891 (0.882,0.901)	0.891 (0.876,0.912)	0.976 (0.97,0.979)	0.977 (0.969,0.99)	0.211 (0.148,0.273)	0.223 (0.143,0.286)
	4.C		FRF	0.891 (0.878,0.897)	0.887 (0.867,0.903)	0.973 (0.969,0.979)	0.975 (0.961,0.99)	0.193 (0.133,0.25)	0.203 (0.125,0.273)
	4.D		MD-BG	0.887 (0.878,0.897)	0.889 (0.867,0.903)	0.97 (0.965,0.975)	0.97 (0.96,0.981)	0.231 (0.176,0.286)	0.244 (0.154,0.333)
Mixed (One Functional + One Non-Functional)	5.A	CCA + Visit	BG	0.908 (0.897,0.916)	0.908 (0.894,0.929)	0.97 (0.965,0.978)	0.97 (0.96,0.989)	0.407 (0.36,0.455)	0.422 (0.308,0.5)
	5.B		RF	0.902 (0.894,0.909)	0.901 (0.885,0.92)	0.981 (0.978,0.987)	0.981 (0.971,0.99)	0.27 (0.219,0.324)	0.284 (0.2,0.364)
	5.C		MD-BG	0.905 (0.897,0.916)	0.906 (0.894,0.92)	0.96 (0.953,0.966)	0.961 (0.949,0.98)	0.467 (0.407,0.531)	0.476 (0.375,0.571)
	5.D		MD-RF	0.887 (0.878,0.897)	0.888 (0.867,0.903)	0.985 (0.97,1)	0.986 (0.971,1)	0.115 (0,0.237)	0.116 (0,0.224)
	6.A	CCA + Gender	BG	0.892 (0.882,0.901)	0.89 (0.876,0.912)	0.97 (0.966,0.975)	0.97 (0.96,0.989)	0.263 (0.206,0.323)	0.264 (0.182,0.333)
	6.B		RF	0.891 (0.882,0.901)	0.89 (0.876,0.903)	0.976 (0.97,0.983)	0.977 (0.97,0.99)	0.214 (0.156,0.269)	0.21 (0.133,0.273)
	6.C		MD-BG	0.887 (0.878,0.897)	0.887 (0.867,0.903)	0.969 (0.962,0.975)	0.97 (0.96,0.981)	0.227 (0.161,0.286)	0.233 (0.154,0.308)
	6.D		MD-RF	0.887 (0.878,0.894)	0.889 (0.876,0.903)	0.984 (0.97,1)	0.985 (0.971,1)	0.12 (0,0.236)	0.117 (0,0.222)
Mixed (One Functional + Two Non-Functional)	7.A	CCA + Visit + Gender	BG	0.907 (0.901,0.916)	0.909 (0.894,0.923)	0.97 (0.965,0.978)	0.972 (0.96,0.99)	0.403 (0.346,0.464)	0.421 (0.312,0.5)
	7.B		RF	0.902 (0.894,0.909)	0.9 (0.885,0.92)	0.981 (0.978,0.987)	0.981 (0.971,0.99)	0.266 (0.207,0.333)	0.272 (0.188,0.333)
	7.C		MD-BG	0.904 (0.894,0.916)	0.907 (0.894,0.92)	0.959 (0.952,0.966)	0.961 (0.949,0.979)	0.462 (0.405,0.517)	0.488 (0.375,0.6)
	7.D		MD-RF	0.894 (0.882,0.905)	0.893 (0.876,0.912)	0.977 (0.962,1)	0.978 (0.961,1)	0.223 (0,0.394)	0.23 (0,0.4)

Response: Patients Status (Categorical) – Algorithms: Bagging (BG) – Mixed Bagging (MD-BG) – Random Forest (RF) – Mixed Random Forest (MD-RF) - Functional Random Forest- fda.usc (FRF) – Number of tree : 1,000 – Number of iterations : 1,000 – Positive Class : MS Patients (Code = 1)

Covariates Type	Model	Covariates	Algorithm	MSPE		COR	
				Train	Test	Train	Test
Non-Functional	1.A	Water	BG	4.244 (3.89,4.607)	4.209 (3.519,4.827)	0.974 (0.971,0.976)	0.973 (0.969,0.979)
	2.A	Protein	BG	53.788 (49.912,58.098)	53.515 (43.863,60.754)	0.666 (0.636,0.693)	0.662 (0.613,0.726)
	3.A	Water + Protein	BG	2.669 (2.452,2.885)	2.648 (2.152,3.062)	0.983 (0.982,0.985)	0.983 (0.98,0.987)
	3.B	Water + Protein	RF	3.297 (3.068,3.511)	3.154 (2.477,3.609)	0.979 (0.978,0.981)	0.98 (0.977,0.985)
Functional	4.A	NIT	BG	62.288 (59.094,65.621)	60.583 (51.749,68.362)	0.613 (0.589,0.638)	0.62 (0.582,0.677)
	4.B	NIT	RF	64.613 (61.335,68)	64.286 (54.586,72.394)	0.598 (0.575,0.623)	0.598 (0.548,0.658)
	4.C	NIT	MBG	72.131 (64.128,78.981)	70.859 (57.344,81.605)	0.552 (0.506,0.606)	0.556 (0.489,0.638)
Mixed (One Functional + One Non-Functional)	5.A	NIT + Water	BG	3.444 (3.198,3.695)	3.367 (2.808,3.877)	0.978 (0.977,0.98)	0.979 (0.976,0.983)
	5.B	NIT + Water	RF	34.387 (32.706,36.128)	33.421 (27.391,37.971)	0.786 (0.774,0.8)	0.791 (0.764,0.826)
	5.C	NIT + Water	MBG	3.453 (3.197,3.714)	3.423 (2.861,3.94)	0.979 (0.977,0.98)	0.978 (0.975,0.983)
	5.D	NIT + Water	MRF	3.465 (3.247,3.687)	3.392 (2.834,3.859)	0.978 (0.977,0.98)	0.979 (0.975,0.983)
	6.A	NIT + Protein	BG	25 (22.782,27.286)	24.87 (19.582,28.815)	0.845 (0.83,0.859)	0.844 (0.82,0.878)
	6.B	NIT + Protein	RF	41.342 (39.326,43.437)	40.197 (32.895,46.301)	0.743 (0.728,0.76)	0.749 (0.717,0.789)
	6.C	NIT + Protein	MBG	25.524 (23.529,27.78)	25.377 (20.232,29.118)	0.841 (0.826,0.855)	0.84 (0.815,0.876)
	6.D	NIT + Protein	MRF	38.067 (4.289,69.359)	37.294 (4.116,68.031)	0.763 (0.563,0.973)	0.769 (0.581,0.975)
Mixed (One Functional + Two Non-Functional)	7.A	NIT + Water + Protein	BG	2.529 (2.321,2.736)	2.494 (2.011,2.929)	0.984 (0.983,0.986)	0.984 (0.981,0.988)
	7.B	NIT + Water + Protein	RF	23.492 (22.358,24.63)	22.819 (18.289,26.585)	0.854 (0.846,0.863)	0.858 (0.839,0.88)
	7.C	NIT + Water + Protein	MBG	2.543 (2.347,2.747)	2.491 (1.993,2.916)	0.984 (0.983,0.986)	0.984 (0.981,0.988)
	7.D	NIT + Water + Protein	MRF	42.604 (4.591,63.584)	41.633 (4.806,64.699)	0.736 (0.604,0.971)	0.735 (0.589,0.969)

Response: Fat (Continuous) – Algorithms: Bagging (BG) – Mixed Bagging (MD-BG) – Random Forest (RF) – Mixed Random Forest (MD-RF) - Number of tree : 1,000 – Number of iterations : 1,000